

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE
ANSWER KEY
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	B	C	B	A	A	A	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	C	A	A	B	B	D	B	C

PART-2 : CHEMISTRY

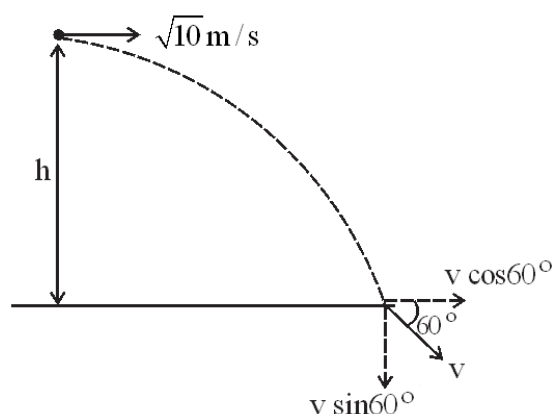
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	C	C	D	B	C	C	A	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	A	B	C	B	D	A	C	D

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	D	A	C	A	D	A	A	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	A	C	D	B	B	A	A	A

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

 1. **Ans (B)**

 Minimum angle of 60° is possible when projectile is thrown horizontally

 $\therefore$ Horizontal velocity remains constant in projectile motion

$$\therefore \sqrt{10} = V \cos 60^\circ$$

$$\Rightarrow V = 2\sqrt{10}$$

Also, vertical motion is free fall motion

$$\therefore \sqrt{2gh} = V \sin 60^\circ$$

$$\Rightarrow \sqrt{20h} = 2\sqrt{10} \times \frac{\sqrt{3}}{2}$$

$$\Rightarrow \sqrt{20h} = \sqrt{30}$$

$$\Rightarrow h = 1.5 \text{ m}$$

2. **Ans (A)**

For ball, $I = \frac{2}{5}mR^2$ (assuming solid sphere)

In case of pure rolling on incline,

$$N = mg \cos \theta$$

$$\text{and } f = \frac{mg \sin \theta}{1 + \frac{I}{mR^2}} = \frac{mg \sin \theta}{1 + \frac{2}{5}} = \frac{5}{7}mg \sin \theta$$

Net force applied on incline

$$\begin{aligned} &= \sqrt{N^2 + f^2} \\ &= \sqrt{m^2 g^2 \cos^2 \theta + \frac{25}{49} m^2 g^2 \sin^2 \theta} \\ &= mg \sqrt{\cos^2 \theta + \sin^2 \theta - \frac{24}{49} \sin^2 \theta} \\ &= mg \sqrt{1 - \frac{24}{49} \sin^2 \theta} \end{aligned}$$

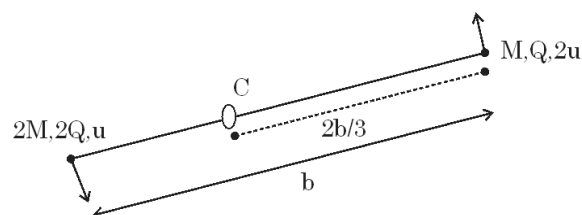
∴ Quantity inside square root is less than one.

∴ This net force is less than mg .

3. **Ans (B)**

In CM frame

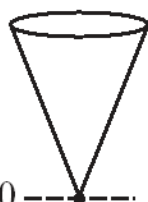
Use linear momentum conservation and angular momentum conservation about CM



4. **Ans (C)**

For hollow cone, COM is at $\frac{2h}{3}$ from vertex

and for solid cone, COM is at $\frac{3h}{4}$ from vertex



$$h = 0$$

$$M_h = 3m$$

$$M_s = 7m - 3m = 4m$$

$$\begin{aligned} \therefore y_{\text{COM}} &= \frac{M_h y_h + M_s y_s}{M_h + M_s} \\ &= \frac{(3m) \left(\frac{2h}{3} \right) + (4m) \left(\frac{3h}{4} \right)}{7m} \\ &= \frac{5Mh}{7m} = \frac{5h}{7} \end{aligned}$$

5. **Ans (B)**

$$\overline{(\bar{A} B) + \bar{C}}$$

$$\begin{aligned} &= \overline{\bar{A} B} \cdot \bar{C} \\ &= \bar{A} \cdot \bar{B} \cdot \bar{C} \end{aligned}$$

6. **Ans (A)**

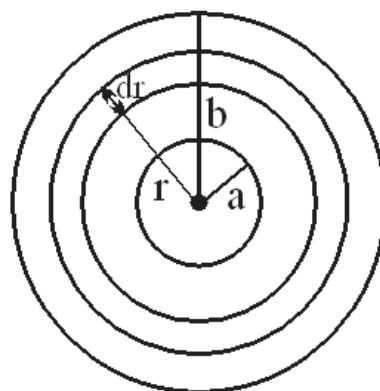
∴ Angular speed of entire wire and that of COM of wire will be same about centre of circular groove. Also, for any point of wire:

$$v = R\omega \Rightarrow \omega = \frac{v}{R}$$

$$\therefore \text{For COM, } \omega = \frac{v}{R}$$

7. **Ans (A)**

$$\int dQ = \int \sigma 2\pi r dr$$



$$Q = \int_a^b \frac{k}{r^3} 2\pi r dr; Q = 2\pi k \int_a^b \frac{dr}{r^2}$$

$$Q = 2\pi k \frac{(b-a)}{ab}; \int dV = \int \frac{k dQ}{r}$$

$$V = \int \frac{1}{4\pi\epsilon_0} \frac{\sigma 2\pi r dr}{r} = \frac{k}{4\pi\epsilon_0} 2\pi \int_a^b \frac{dr}{r^3}$$

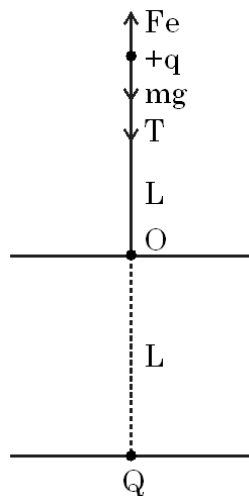
$$= \frac{1}{4\pi\epsilon_0} \frac{k 2\pi}{2} \left(\frac{b^2 - a^2}{a^2 b^2} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2\pi k}{2} \frac{(b-a)}{ab} \frac{(b+a)}{ab}$$

$$= \frac{1}{8\pi\epsilon_0} Q \left(\frac{a+b}{ab} \right)$$

8. Ans (A)

In initial equilibrium position



$$T + mg = Fe$$

$$2mg + mg = Fe$$

$$Fe = 3mg \quad \dots\dots\dots(1)$$

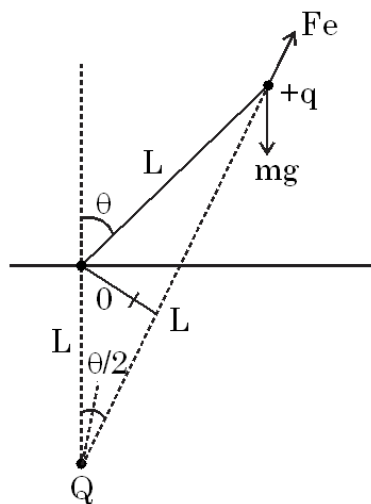
When slightly displaced

We will neglect slight change in Fe as distance b/w charges will change.

$$i_0 = I\alpha$$

$$mg \cdot L \sin \theta - FeL \sin \frac{\theta}{2} = mL^2 \alpha$$

$$\text{From (1) } Fe = 3mg$$



$$mgL\theta - 3mgL \frac{\theta}{2} = mL^2 \alpha$$

$$-\frac{mg}{2}L\theta = mL^2 \alpha$$

$$\alpha = -\left(\frac{g}{2L}\right)$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{2L}{g}}$$

9. Ans (D)

$$R_1 = \frac{d_1}{\sigma_1 A}, R_2 = \frac{d_2}{\sigma_2 A}$$

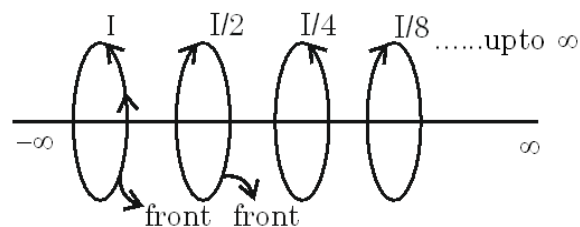
$$\therefore i = \frac{V}{\frac{d_1}{\sigma_1 A} + \frac{d_2}{\sigma_2 A}}$$

$$\therefore j = \frac{i}{A} = \frac{V}{\frac{d_1}{\sigma_1} + \frac{d_2}{\sigma_2}}$$

$$j = \sigma E \quad \therefore E = \frac{j}{\sigma}$$

$$\therefore E_A = \frac{j}{\sigma_1} = \frac{V}{d_1 + \frac{\sigma_1 d_2}{\sigma_2}} = \frac{V \sigma_2}{\sigma_1 d_2 + \sigma_2 d_1}$$

10. Ans (C)



Lets take a square loop of infinite length with one of the sides being the axis of the circular loops itself.

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \left[-I + \frac{I}{2} - \frac{I}{4} + \frac{I}{8} - \dots \right]$$

But apart from the axial side, $\int \vec{B} \cdot d\vec{\ell}$ for all other sides will be zero as magnetic field along those sides is negligible.

$$\therefore \int_{-\infty}^{\infty} \vec{B} \cdot d\vec{x} + 0 + 0 + 0 = -\mu_0 I \left[1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots \right] = -\mu_0 I \left[\frac{1}{1 - \left(\frac{-1}{2}\right)} \right] = \frac{-2\mu_0 I}{3}$$

11. **Ans (C)**

Emf induced in the first quadrant equal to zero.

Emf induced in the second quadrant

$$= \int_{x=0}^{x=R} B_0 (dx) \omega x = \frac{B_0 \omega R^2}{2}$$

$$\text{Emf induced in the third quadrant} = \frac{-3B_0 \omega R^2}{2}$$

Emf induced in the fourth quadrant

$$= \frac{7B_0 \omega R^2}{2}$$

$$\therefore \text{Avg.} = \frac{\frac{B_0 \omega R^2}{2} - \frac{3B_0 \omega R^2}{2} + \frac{7B_0 \omega R^2}{2}}{4} = \frac{5B_0 \omega R^2}{8}$$

12. **Ans (A)**

Initial	550K	250K
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Final	T_{1f}	T_{2f}
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Work done by adiabatically expanding gas is equal and opposite to work done by adiabatically compressing gas.

$$\therefore \frac{nR}{1-\gamma} (T_{1f} - 550) = -\frac{nR}{1-\gamma} (T_{2f} - 250)$$

$$\Rightarrow T_{1f} + T_{2f} = 800 \dots (i)$$

$\therefore PV^\gamma = \text{constant}$ for both sides

$$\therefore P_{1i} V_{1i}^\gamma = P_{1f} V_{1f}^\gamma \text{ and } P_{2i} V_{2i}^\gamma = P_{2f} V_{2f}^\gamma$$

Dividing above equations:

$$\frac{P_{1i} V_{1i}^\gamma}{P_{2i} V_{2i}^\gamma} = \frac{P_{1f} V_{1f}^\gamma}{P_{2f} V_{2f}^\gamma} \Rightarrow \frac{P_{1i}}{P_{2i}} = \left(\frac{V_{1f}}{V_{2f}} \right)^\gamma$$

$$(\because V_{1i} = V_{2i} \text{ \& } P_{1f} = P_{2f})$$

$$\Rightarrow \frac{nR T_{1i} / V_{1i}}{nR T_{2i} / V_{2i}} = \left(\frac{nR T_{1f} / P_{1f}}{nR T_{2f} / P_{2f}} \right)^\gamma$$

$$\Rightarrow \frac{T_{1i}}{T_{2i}} = \left(\frac{T_{1f}}{T_{2f}} \right)^\gamma$$

$$(\because V_{1i} = V_{2i} \text{ \& } P_{1f} = P_{2f})$$

$$\Rightarrow \frac{T_{1f}}{T_{2f}} = \left(\frac{550}{250} \right)^{\frac{1}{(7/5)}} \dots (ii)$$

Solving (i) & (ii),

$$T_{1f} = 510K$$

13. **Ans (C)**

When switch is open, charge on inner surface of shell = $-q$.

$$\text{Charge on outer surface} = 2q - (-q)$$

$$= 3q. \text{ (as total charge on shell} = 2q)$$

When switch is closed, potential of the shell becomes zero.

$V_{\text{shell}} = \text{Work done against electric field in moving a unit charge from infinity to surface}$
 $= \text{Work done against the field of } 2q \text{ (at A), } 2q \text{ (at B) and } q' \text{ (on outer surface of the shell) from infinity to surface}$

Note that the external field will have no contribution from the charges inside the shell due to shielding effect.

$$\therefore V_{\text{shell}} = \text{work done against the field of } 2q, 2q, \text{ \& } q' \text{ from infinity to centre of the shell.}$$

(as field inside shell due to these charges = 0)

= potential at centre due to $2q, 2q$ & q' .

$$\Rightarrow 0 = \frac{k2q}{2r} + \frac{k2q}{2r} + \frac{kq'}{r}$$

$$\Rightarrow 0 = q + q + q' \therefore q' = -2q$$

$$\therefore \text{Charge flow} = 3q - (q')$$

$$= 3q - (-2q) = 5q$$

14. **Ans (A)**

$$T_{\text{mixture}} = \frac{m_1 s_1 T_1 + m_2 s_2 T_2 + m_3 s_3 T_3}{m_1 s_1 + m_2 s_2 + m_3 s_3}$$

Substituting the values

$$T_{\text{mixture}} = 23.6C$$

15. **Ans (A)**

Snell's Law at O

$$\mu \sin \theta = 1 \sin \phi$$

for small angle

$$\sin \theta \simeq \tan \theta$$

$$\sin \phi \simeq \tan \phi$$

$$\mu \tan \theta = \tan \phi$$

$$\mu \frac{h}{R+x} = \frac{h}{x}$$

$$\mu x = R+x$$

$$x = \frac{R}{\mu - 1}$$

16. **Ans (B)**

$${}_Z^AY \rightarrow {}_{Z-1}^AX + {}_{+1}^0\beta + \nu$$

$$Q = [(m_Y - zm_e) - (m_x - (z-1)m_e) - m_e]c^2$$

$$\Rightarrow Q = (m_Y - m_x - 2m_e)c^2$$

For reaction to be feasible $Q > 0$

$$\Rightarrow m_Y - m_x - 2m_e > 0$$

$$\Rightarrow m_Y - m_x > 2m_e$$

17. **Ans (B)**

$$N = N_s + N_u \quad (1)$$

$$\frac{N}{3} = N_s + \frac{N_u}{x} \quad (2)$$

Divide equation (1) by equation (2)

$$3 = \frac{N_s + N_u}{N_s + \frac{N_u}{x}}; 3 = \frac{N_s/N_u + 1}{N_s/N_u + \frac{1}{x}}$$

$$\text{Let } \frac{N_s}{N_u} = t; 3 = \frac{t+1}{t+\frac{1}{x}}$$

$$3t + \frac{3}{x} = t + 1$$

$$2t = 1 - \frac{3}{x} \Rightarrow t = \frac{x-3}{2x}$$

18. **Ans (D)**

In sample x no impurity level seen, so it is undoped. In sample y impurity energy level lies closer to the conduction band so it is doped with fifth group impurity.

In sample z, impurity energy level lies closer to the valence band so it is doped with third group impurity.

19. **Ans (B)**

Susceptibility χ for a paramagnetic substance varies with absolute temperature as $\chi \propto \frac{1}{T}$

$$\therefore \frac{\chi_2}{\chi_1} = \frac{T_1}{T_2}$$

$$\therefore \chi_2 = \frac{300}{350} \times 2.5 \times 10^{-5}$$

$$= 2.14 \times 10^{-5}$$

$\therefore$ Magnetisation at 350 K is

$$I = \chi H = 2.14 \times 10^{-5} \times 2000 \text{ Am}^{-1}$$

$$= 4.28 \times 10^{-2} \text{ Am}^{-1}$$

20. **Ans (C)**

Using energy conservation

$$mg \frac{L}{2} (1 - \cos \theta) = \frac{1}{2} \left(\frac{ML^2}{3} \right) \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{3g}{5L}} = \sqrt{\frac{30}{5 \times 1.5}} = 2 \text{ rad/s}$$

Now, at final position, consider an elemental mass dm having width dx at a distance x from end A.

Linear momentum of this element:

$$dP = (dm) v$$

$$= \left(\frac{M}{L} dx \right) (\omega x) = \frac{M}{L} \omega x dx$$

$$\therefore P = \int dP = \frac{M}{L} \omega \int_0^L x dx$$

$$= \frac{M\omega}{L} \left(\frac{L^2}{2} \right) = \frac{M\omega L}{2}$$

$$= \frac{5 \times 2 \times 1.5}{2} = 7.5 \text{ kg m/s}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (12)**

$$i = \epsilon_0 A \frac{dE}{dt}$$

2. **Ans (6)**

Using Bernoulli's theorem at surface of water-gas boundary and just outside nozzle:

$$P_{\text{gas}} = P_{\text{atm}} + \frac{1}{2} \rho v^2$$

[Neglecting pressure due to height of water & also neglecting speed of water-gas boundary]

$$\Rightarrow 4 \times 10^6 = 1 \times 10^6 + \frac{1}{2} \times 10^3 v^2$$

$$\Rightarrow v^2 = 6 \times 10^3$$

$$\therefore n = 6$$

3. Ans (2)

$$\tan 30^\circ = \frac{B_V}{B_H + B_R} = \frac{1}{\sqrt{3}}$$

$$\tan 60^\circ = \frac{B_V}{B_H - B_R} = \sqrt{3}$$

Dividing, we get

$$\frac{B_H - B_R}{B_H + B_R} = \frac{1}{3} \Rightarrow 3B_H - 3B_R = B_H + B_R$$

$$\Rightarrow 2B_H = 4B_R \therefore B_R = \frac{B_H}{2}$$

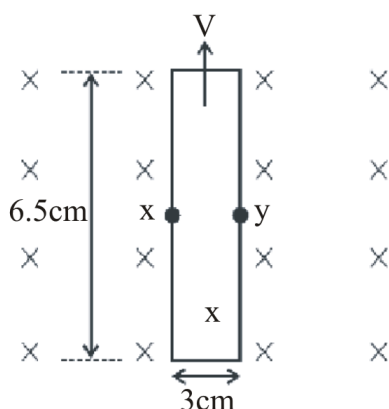
$$50 \frac{\mu_0 i}{2R} = \frac{B_H}{2}$$

$$\therefore R = \frac{\mu_0 i \times 50}{B_H} = \frac{(4\pi \times 10^{-7}) \times 3 \times 50}{4 \times 3 \times 10^{-5}}$$

$$= \pi \times 10^{-2} \times 50 \text{ m}$$

$$k = 2$$

4. Ans (10)

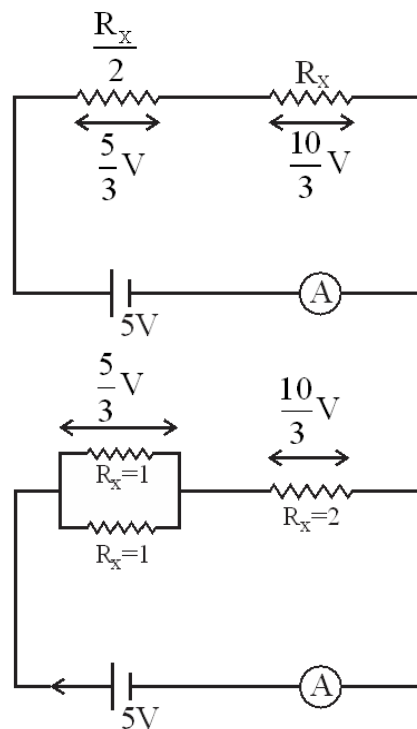


$$\text{emf} = Blv$$

$$\Rightarrow 3.6 \times 10^{-6} = (1.2 \times 10^{-3}) (3 \times 10^{-2}) v$$

$$\Rightarrow v = 10 \text{ cm/s}$$

5. Ans (2)



$$i = \frac{5}{2 + \frac{1}{2}} = 2 \text{ A}$$

6. Ans (1)

$\therefore$ In collision, V_{COM} just before & just after collision remains same.

$$\therefore V_{\text{COM}i} = \frac{mv + 0}{2m} = \frac{v}{2} = \frac{2}{2} = 1$$

$$\text{So, } V_{\text{COM}f} = V_{\text{COM}i} = 1 \text{ m/s}$$

7. Ans (8)

$m = -d_i/d_0$. Solving this expression for d_i gives

$$d_i = -md_0$$

Entering known values yields

$$d_i = -(0.0320)(12.0 \text{ cm}) = -0.384 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{d_0} + \frac{1}{d_i}$$

Substituting known values,

$$\frac{1}{f} = \frac{1}{12.0 \text{ cm}} + \frac{1}{-0.384 \text{ cm}} = \frac{-2.52}{\text{cm}}$$

$$f = \frac{\text{cm}}{-2.52} = -0.400 \text{ cm}$$

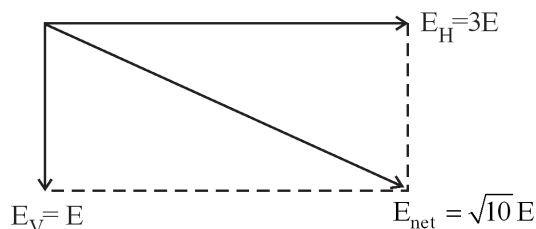
8. Ans (72)

$$\therefore \text{COP} = \frac{T_C}{T_H - T_C} = \frac{4}{292 - 4} = \frac{4}{288}$$

$$\text{Also, } \text{COP} = \frac{Q_C}{W}$$

$$\therefore W = \frac{Q_C}{\text{COP}} = \frac{1}{4/288} = 72 \text{ J}$$

9. Ans (1)



$\therefore$ Energy density $\propto E_{\text{net}}^2$

$$\therefore \frac{\text{Energy After}}{\text{Energy Before}} = \frac{(E)^2}{(\sqrt{10}E)^2} = 0.1$$

10. Ans (1)

In time interval of 0.2s, distance travelled by pulse = $10 \times 0.2 = 2$ cm

This means the point $x = 2$ cm at $t = 0.2$ s was at $x = 0$ at $t = 0$.

$\therefore$ At $t = 0$, $y = 1$ cm for $x = 0$

$\therefore y = 1$ cm for $x = 2$ cm at $t = 0.2$ s.

PART-2 : CHEMISTRY

SECTION-I

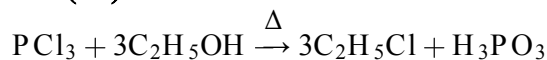
1. Ans (B)

Correct order of EA = $\text{Cl} > \text{F} > \text{Br} > \text{I}$

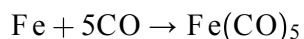
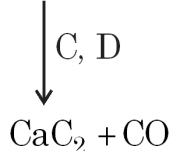
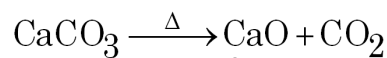
2. Ans (D)

Due to high polarising power of Be^{+2} , BeCO_3 is thermally less stable.

4. Ans (C)



5. Ans (D)

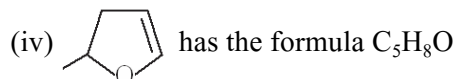
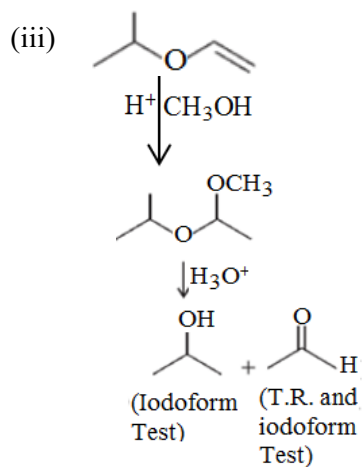
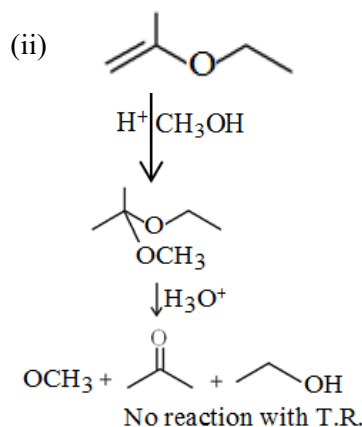
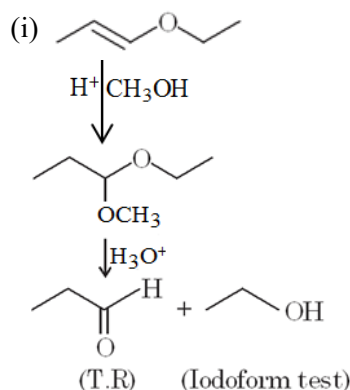


7. Ans (C)

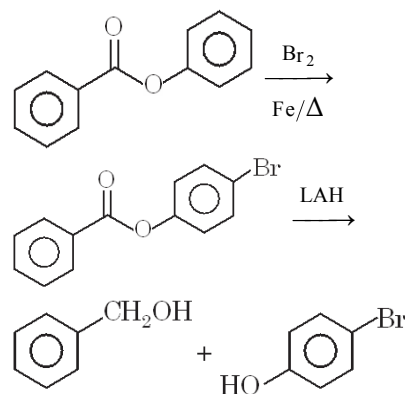
Molarity of 10V $\text{H}_2\text{O}_2 = 0.8$ gm

$$\% \frac{w}{w} = \frac{0.89 \times 34}{1000} \times 100 = 3\%$$

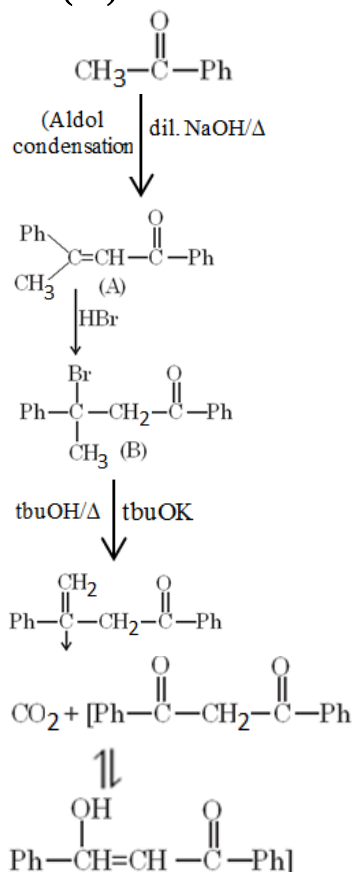
8. Ans (C)



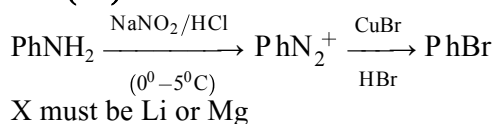
9. Ans (A)



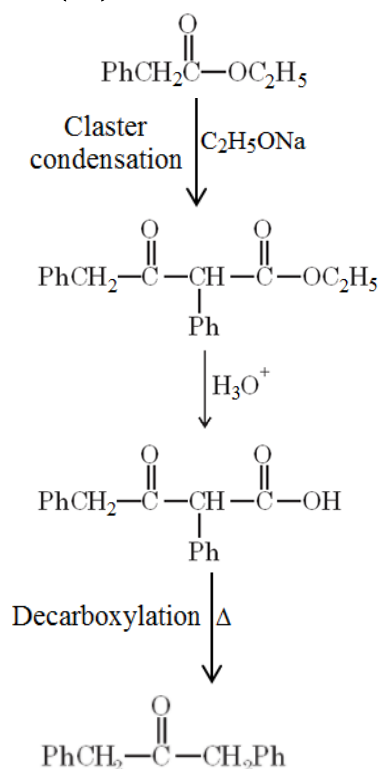
10. Ans (C)



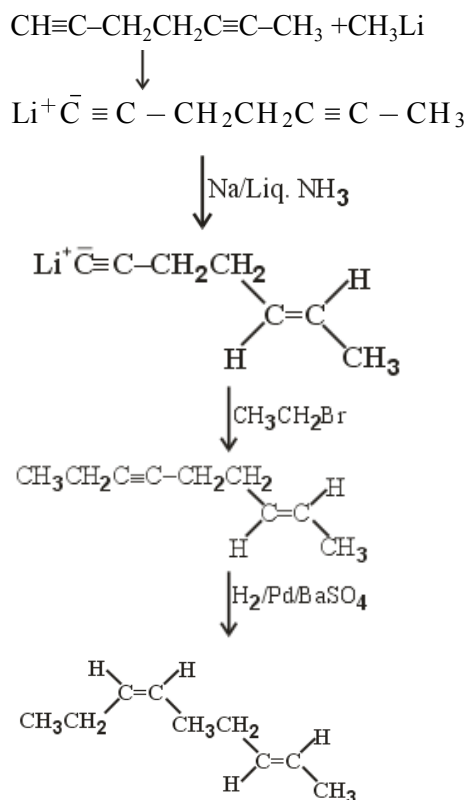
11. Ans (B)



12. Ans (C)

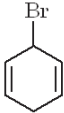
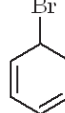


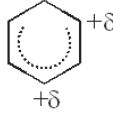
13. Ans (A)



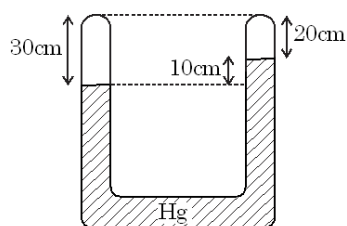
14. Ans (B)

The most stable intermediate carbocation is

formed from both  and  that is

. However (B) is less stable, and hence more reactive.

15. Ans (C)



For air : $P_1V_1 = P_2V_2$

$$\Rightarrow 80 \times 25 = P_2 \times 20$$

$$\Rightarrow P_2 = 100 \text{ cm for Hg}$$

For the left column :-

$$P_{\text{left}} = P_{\text{right}} + 10 \text{ cm of Hg}$$

$$\Rightarrow P_{\text{left}} = 110 \text{ cm of Hg}$$

Also, as T is constant

$$\frac{P_1V_1}{n_1R} = \frac{P_2V_2}{n_2R}$$

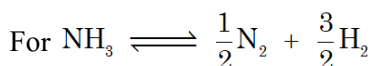
Where,

n_1 = moles before dissociation

n_2 = moles after dissociation

$$\Rightarrow \frac{80 \times 25}{n_1} = \frac{110 \times 30}{n_2}$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{33}{20} = 1.65 \quad (1)$$



the n_1

eqn

$$n_1(1 - \alpha)$$

$$\frac{n_1 \alpha}{2}$$

$$\frac{3n_1 \alpha}{2}$$

$$\therefore n_2 = n_1(1 - \alpha) + \frac{n_1 \alpha}{2} + \frac{3n_1 \alpha}{2}$$

$$\Rightarrow n_2 = n_1(1 + \alpha)$$

$$\Rightarrow \frac{n_2}{n_1} = 1 + \alpha = 1.65$$

$$\Rightarrow \alpha = 0.65$$

16. Ans (B)

Since O_2 is added at constant partial pressure, the total pressure decreases and the total volume increases.



Let partial pressure x y z

at initial eq1 be

$$\therefore K_p = \frac{y^4 z}{x^2} \quad \dots(1)$$

Pressure after

adding O_2 , x/a y/a z

assuming that

volume increases

to 'a' times the initial

volume

$$\therefore Q_p = \frac{(y/a)^4 z}{(x/a)^2} = \frac{y^4 z}{x^2} \times \frac{1}{a^2} \quad \dots(2)$$

$$\therefore Q_p < K_p \quad \text{equilibrium shift fowards.}$$

17. Ans (D)

$$\Delta U = W$$

$$\Rightarrow \frac{P_2 V_2 - P_1 V_1}{\gamma - 1} = -P_{\text{ext}} \Delta V$$

$$\Rightarrow \frac{P_{\text{ext}} V_2 - 20}{5/3 - 1} = -P_{\text{ext}} (V_2 - 20)$$

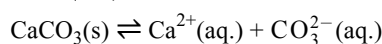
$$\Rightarrow \frac{P_{\text{ext}} V_2 - 20}{2/3} = -P_{\text{ext}} V_2 + 20 P_{\text{ext}}$$

$$\Rightarrow 3P_{\text{ext}} V_2 - 60 = -2P_{\text{ext}} V_2 + 40P_{\text{ext}}$$

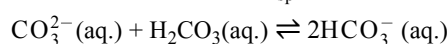
$$\Rightarrow V_2 = \frac{60 + 40 P_{\text{ext}}}{5 P_{\text{ext}}} = \frac{12}{P_{\text{ext}}} + 8$$

$$V_{2\text{min}} = 8\text{L (when } P_{\text{ext}} \text{ is max}^m)$$

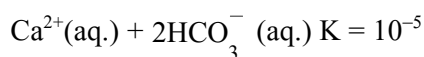
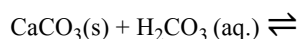
18. Ans (A)



$$K_{\text{sp}} = 10^{-9}$$



$$K_c = \frac{K_1}{K_2} = 10^4$$



$$\begin{array}{ccc} \text{in} & 0.4 & - \\ \text{eq}^1 & (0.4-x) & x \end{array} \quad \begin{array}{ccc} - & - & - \\ & & 2x \end{array}$$

$$K = \frac{[\text{Ca}^{2+}][\text{HCO}_3^{-}]^2}{[\text{H}_2\text{CO}_3]}$$

$$\Rightarrow 10^{-5} = \frac{x \cdot (2x)^2}{(0.4-x)} \simeq \frac{4x^3}{0.4}$$

$$\Rightarrow x^3 = 10^{-6}$$

$$\Rightarrow x = 10^{-2} \text{ mol L}^{-1}$$

$$= 1 \text{ g CaCO}_3 \text{ per L of sol}^n.$$

19. Ans (C)

(i) $\therefore -\frac{d[A]}{dt} = k$ (from the graph), it is a zero order reaction.

(ii) unit of rate constant for a zero order reaction is $\text{mol L}^{-1}\text{s}^{-1}$

(iii) Rate constant does not depend on concentration of the reaction

(iv) Zero order reaction is always a complex reaction.

20. **Ans (D)**
For an e^- in a Bohr's orbit,
 $2\pi r = n\lambda$

$$\Rightarrow 2\pi r_0 \frac{n^2}{Z} = n\lambda \quad \Rightarrow \lambda = 2\pi r_0 \frac{n}{Z}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{n_2 Z_1}{n_1 Z_2} \quad \Rightarrow \frac{10}{1} = \frac{n_2 \times 2}{1 \times 1}$$

$$\Rightarrow n_2 = 5$$

Upon de-excitation visible lines in Balmer series corresponds to the transitions

$$5 \rightarrow 2$$

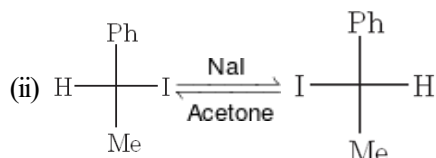
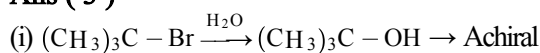
$$4 \rightarrow 2$$

$$3 \rightarrow 2$$

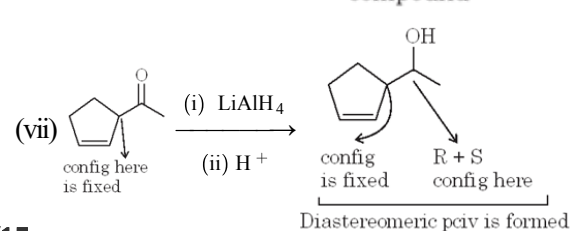
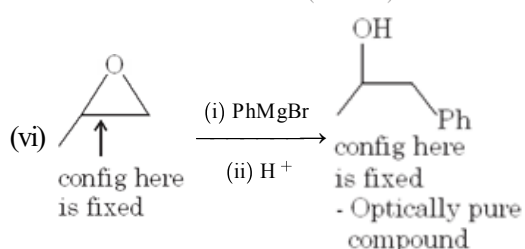
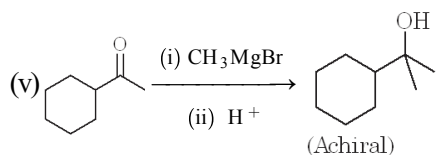
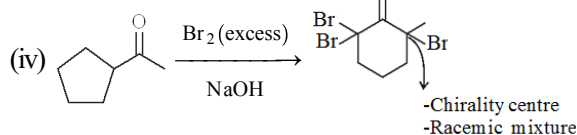
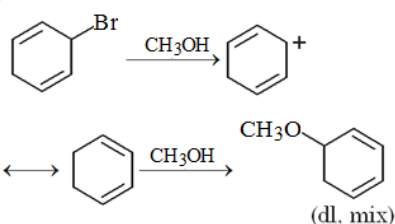
PART-2 : CHEMISTRY

SECTION-II

4. **Ans (3)**



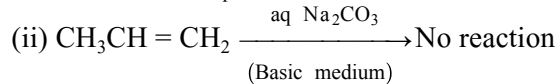
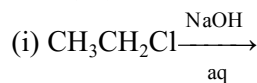
(iii)



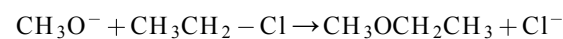
5. **Ans (5)**

(ii), (iv), (v), (vi) and (x)

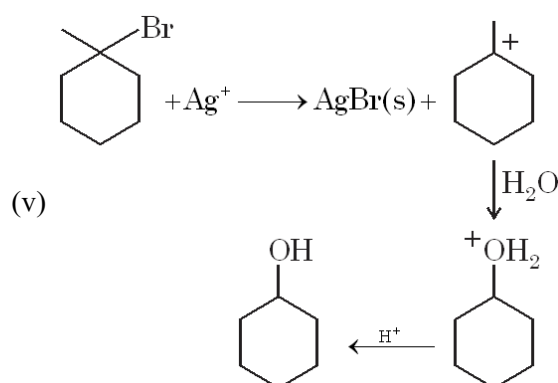
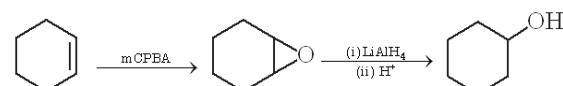
6. **Ans (4)**



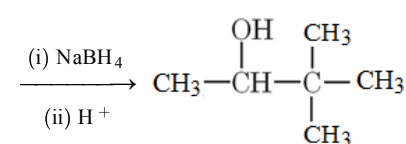
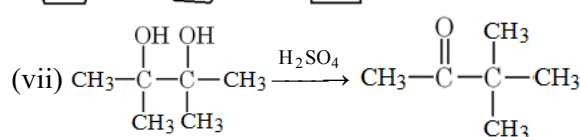
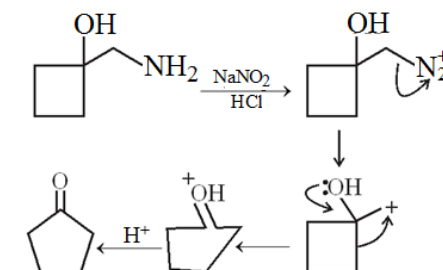
(iii)



(iv)

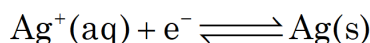


(v)



(Pinacol Pinacolone rearrangement)

7. **Ans (37)**



$$E_{\text{red}} = E^0 - \frac{RT}{nF} \log \frac{1}{[\text{Ag}^+]}$$

$$\Rightarrow E_{\text{red}} = E^0 + \frac{RT}{F} \log [\text{Ag}^+]$$

$$\therefore \text{Slope of } E_{\text{red}} \text{ vs } \log [\text{Ag}^+], M = \frac{RT}{F}$$

Given $\frac{RT}{F} = 0.06$ at 300K and 0.062 at the temperature of the solution.

$$\Rightarrow \frac{RT_1/F}{RT_2/F} = \frac{0.06}{0.062}$$

$$\Rightarrow T_2 = \frac{0.062}{0.06} \times T_1$$

$$= \frac{0.062}{0.06} \times 300 = 310\text{K} = 37^\circ\text{C}$$

8. **Ans (8)**

$$\frac{r_+}{r_-} = \frac{1.4}{2.6} = 0.538$$

$$\Rightarrow \text{C.N.} = 6$$

The unit cell must be NaCl type

$$\therefore \text{Edge length} = 2(r_+ + r_-) = 8 \text{ \AA}$$

9. **Ans (25)**

Isotonic solutions have same molarity

Let molar mass of solute S = M

Concentration = 0.5 g solute in 100 ml solution

= 5 g solute in 1000 ml solution

$$= \frac{5}{M} \text{ moles S in 1 L solution}$$

$$\text{Molarity} = \frac{5}{M} = 0.2 \text{ (given)}$$

$$\Rightarrow M = 25 \text{ g/mol}$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

Let us assume that no. of cups with handle is n and without handle is m. No. of ways of selecting one cup each is ${}^mC_1 \cdot {}^nC_1 = 36$, $mn = 36$
Now apply A. M. $\geq$ G. M.

$$\frac{m+n}{2} \geq \sqrt{mn}$$

$$\frac{m+n}{2} \geq 6$$

$$m+n \geq 12$$

So least value is '12'

2. **Ans (C)**

$$(x+10)(x-a) = -5$$

Case (i) Either $x+10=5$, $x-a=-1$

$$x=5, -5-a=-1$$

$$a=-4$$

$$\text{or } x+10=-1, x-a=5$$

$$x=-11, -11-a=5$$

$$a=-16$$

Case (ii) $x+10=-5$, $x-a=1$

or $x+10=1$, $x-a=-5$ it gives same values of 'a'.

Sum of values of 'a' is $-16-4=-20$

3. **Ans (D)**

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1; x, y, z > 0$$

$$\text{Let } x = \frac{1}{a}, y = \frac{1}{b}, z = \frac{1}{c} \Rightarrow a+b+c=1$$

$$\text{Also } (x-1)(y-1)(z-1)$$

$$= \left(\frac{1}{a}-1\right) \left(\frac{1}{b}-1\right) \left(\frac{1}{c}-1\right)$$

$$= \frac{(1-a)(1-b)(1-c)}{abc}$$

$$= \frac{(b+c)(a+c)(a+b)}{abc}$$

Now by A. M. $\geq$ G. M.

$$\frac{b+c}{2} \geq \sqrt{bc}$$

$$b+c \geq 2\sqrt{bc}$$

$$c+a \geq 2\sqrt{ac}$$

$$a+b \geq 2\sqrt{ab}$$

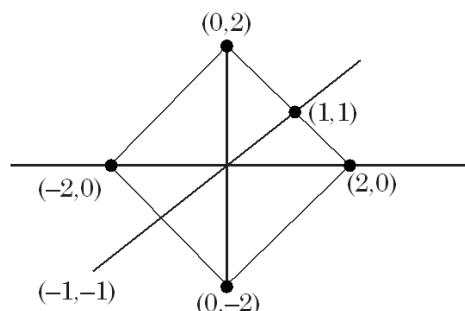
$$(b+c)(c+a)(a+b) \geq 8abc$$

$$\frac{(b+c)(c+a)(a+b)}{abc} \geq 8$$

4. Ans (A)

$$y = x + \cos x$$

$$\frac{dy}{dx} = 1 - \sin x = 0 \Rightarrow \sin x = 1 \Rightarrow x = 2n\pi + \frac{\pi}{2}$$



Putting value of x in given function

$$y = x + \cos x$$

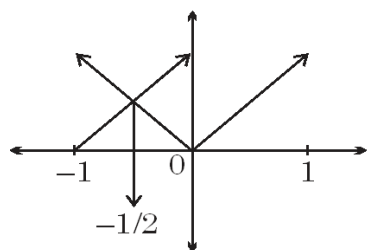
$$y = 2n\pi + \frac{\pi}{2} + 0$$

$$y = 2n\pi + \frac{\pi}{2}$$

Clearly point lie on the line $y = x$

Now x can be between -1 and 1 whereas least value of x is $\frac{\pi}{2}$, if $x > 0$ max value of x is $-\frac{\pi}{2}$ if $x < 0$. So no value of ' x '.

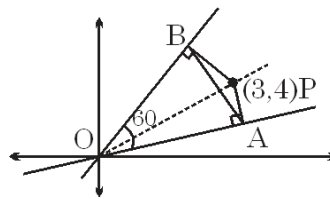
5. Ans (C)



$$\text{Clearly, } f(x) = \begin{cases} -x, & -1 \leq x < -\frac{1}{2} \\ x+1, & -\frac{1}{2} \leq x < 0 \\ x, & 0 \leq x \leq 1 \end{cases}$$

$$\begin{aligned} & \int_{-1}^{-1/2} -x \cdot dx + \int_{-1/2}^0 (x+1) dx + \int_0^1 x \cdot dx \\ &= -\frac{1}{2} [x^2]_{-1}^{-1/2} + \left[\frac{x^2}{2} + x \right]_{-1/2}^0 + \left[\frac{x^2}{2} \right]_0^1 \\ &= -\frac{1}{2} \left[\frac{1}{4} - 1 \right] + \left[0 - \left(\frac{1}{8} - \frac{1}{2} \right) \right] + \frac{1}{2} \\ &= \frac{3}{8} + \left(\frac{3}{8} \right) + \frac{1}{2} = \frac{3}{4} + \frac{1}{2} = \frac{5}{4} \end{aligned}$$

6. Ans (A)

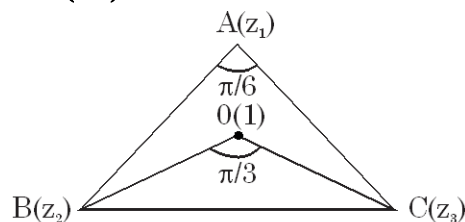


Apply sin rule in triangle OAB, $\frac{AB}{\sin 60^\circ} = OP$

$$OP = \sqrt{3^2 + 4^2} = 5$$

$$AB = 5 \cdot \sin 60^\circ = \frac{5\sqrt{3}}{2}$$

7. Ans (D)



$z = 1$ is circum center. So triangle OBC is equilateral

$$z_2^2 + z_3^2 + 1 = z_2 + z_3 + z_2 z_3$$

$$\text{Expression } z_2^2 - z_2 - z_2 z_3 - z_3 + z_3^2 + 1 = 0$$

$$z_2(z_2 - 1) - z_3(z_2 + 1) + (z_3 + 1)(z_3 - 1) = -2$$

So value is -2

8. Ans (A)

$$1 + |\sin \theta| + |\cos \theta| - 1 - 2 = 0$$

$$\Rightarrow |\sin \theta| + |\cos \theta| = 2 \text{ which is not possible.}$$

9. Ans (A)

Divide throughout by $x^3 y$, we get

$$2 \frac{dx}{x} + 2 \frac{dy}{y} - 2 \frac{y^3}{x^3} dx + 3 \frac{y^2}{x^2} dy = 0$$

$$\Rightarrow 2 \frac{dx}{x} + 2 \frac{dy}{y} + \frac{3xy^2 dy - 2y^3 dx}{x^3} = 0$$

$$\Rightarrow 2 \frac{dx}{x} + 2 \frac{dy}{y} + \frac{3x^2 y^2 dy - 2xy^3 dx}{x^4} = 0$$

$$\Rightarrow 2 \frac{dx}{x} + 2 \frac{dy}{y} + d \left(\frac{y^3}{x^2} \right) = 0, \text{ integrate}$$

$$\Rightarrow 2(\ln x + \ln y) + \frac{y^3}{x^2} = C, \text{ Now put } x = y = 1$$

$$\Rightarrow 2 \ln(1) + 1 = C \Rightarrow C = 1 \Rightarrow 2 \ln(xy) + \frac{y^3}{x^2} = 1$$

$$m = 3, n = 2 \Rightarrow m + n = 5$$

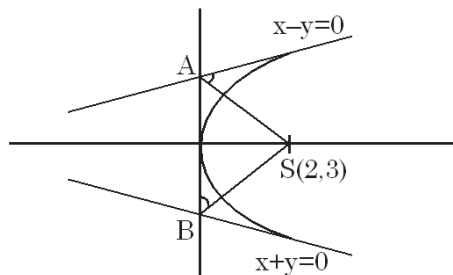
10. **Ans (B)**

Minimum value of $\sqrt{(x-1)^2 + (y-1)^2}$

Clearly minimum distance of circle from the point (1, 1)

$$\sqrt{(4-1)^2 + (5-1)^2} = 5, 7-5 = 2$$

11. **Ans (A)**



Equation of AS :

$$(y-3) = -1(x-2)$$

$$y-3 = -x+2$$

$$x+y=5$$

For point 'A' solve

$$x+y=5$$

$$y=x$$

$$2x=5$$

$$x = \frac{5}{2}, y = \frac{5}{2}$$

$$A \left(\frac{5}{2}, \frac{5}{2} \right)$$

Equation of BS :

$$(y-3) = 1(x-2)$$

$$y-3 = x-2$$

$$x-y=-1$$

For point 'B' solve

$$x-y=-1$$

$$x+y=0$$

$$2x=-1$$

$$x = -\frac{1}{2}, y = \frac{1}{2}$$

Now equation of line AB

$$y - \frac{1}{2} = \frac{\frac{5}{2} - \frac{1}{2}}{\frac{5}{2} + \frac{1}{2}} \left(x + \frac{1}{2} \right)$$

$$y - \frac{1}{2} = \frac{2}{3} \left(x + \frac{1}{2} \right)$$

$$3y - \frac{3}{2} = 2x + 1$$

$$6y - 3 = 4x + 2$$

$$4x - 6y + 5 = 0$$

Option 'A' is correct.

12. **Ans (D)**

$$S(6, 13) \text{ and } S'(25, 8)$$

$$P(1, 1), S'P = 25, SP = 13$$

$$S'P - SP = 12$$

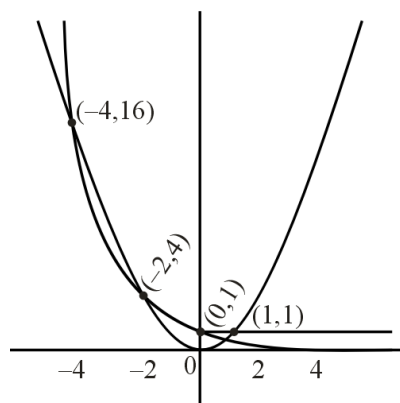
$$2a = 12$$

$$S'S = \sqrt{(25-6)^2 + (8-13)^2}$$

$$= \sqrt{361 + 25} = \sqrt{386} = 2ae$$

$$e = \frac{\sqrt{386}}{12}$$

13. **Ans (A)**



From graph clearly '4' points of non-differentiability.

14. **Ans (C)**

For $x = 1$ $0 = f(y) + f\left(\frac{1}{y}\right)$, interchanging x

and y , we have $2f(y) = f(xy) + f\left(\frac{y}{x}\right)$ adding

$$f(xy) = f(x) + f(y)$$

Partial differentiation with respect to 'x'

$$f'(xy) \cdot y = f'(x) \dots\dots\dots(1)$$

Partial differentiation with respect to 'y'

$$f'(xy) \cdot x = f'(y) \dots\dots\dots(2)$$

$$\frac{y}{x} = \frac{f'(x)}{f'(1)} \text{ put } y = 1$$

$$\frac{1}{x} = \frac{f'(x)}{f'(1)}$$

$$\frac{1}{\ln 6} \times \frac{1}{x} = f'(x)$$

Integrate with respect to 'x'

$$f(x) = \frac{1}{\ln 6} \cdot \ln x + C \text{ put } x = 1$$

$$f(1) = 0 + C = 0$$

$$f(x) = \frac{\ln x}{\ln 6}$$

$$f(7776) = \frac{\ln(7776)}{\ln 6} = 5$$

15. **Ans (D)**

$$((\vec{b} \times \vec{c}) \cdot \vec{a})\vec{c} - ((\vec{b} \times \vec{c}) \cdot \vec{c})\vec{a} = 3\vec{c}$$

$$[\vec{a} \ \vec{b} \ \vec{c}]\vec{c} = 3\vec{c}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = 3$$

$$\text{We know that } [\vec{b} \times \vec{c} \ \vec{c} \times \vec{a} \ \vec{a} \times \vec{b}] = [\vec{a} \ \vec{b} \ \vec{c}]^2 = 9$$

16. **Ans (B)**

There will be no effect due to addition of constant in variables.

17. **Ans (B)**

$$\begin{bmatrix} 0 & \otimes \\ -\otimes & 0 \end{bmatrix} \text{ If matrix is of order } 2 \times 2, \text{ then } \otimes$$

this place can be filled in 6 ways so 6 matrices can be formed. If matrix is of order 3×3 , then

$$\begin{bmatrix} 0 & \otimes & \ominus \\ -\otimes & 0 & \oplus \\ -\ominus & -\oplus & 0 \end{bmatrix}$$

$\otimes$ this place can be filled in 6 ways

$\ominus$ this place can be filled in 4 ways

$\oplus$ this place can be filled in 2 ways

So total matrices $6 \times 4 \times 2 = 48$

Now total matrices $48 + 6 = 54$

18. **Ans (A)**

$$\sum_{r=0}^m r \cdot r! = ((r+1) - 1)r! = (r+1)! - r!$$

$$\sum_{r=0}^m (r+1)! - r! = (m+1)! - 1$$

$$({}^m C_0)^2 + ({}^m C_1)^2 + ({}^m C_2)^2 + \dots + ({}^m C_m)^2 = {}^{2m} C_m$$

$$\sum_{r=0}^m (2r-1) = m^2 - 1$$

So Row 1 and Row 3 is same so value of determinant is zero.

19. **Ans (A)**

Putting (0, 0, 0) expression is -1 and putting

(1, 1, 1) expression is $2 + a^2 + ab - 1$

Both should have opposite sign so

$$2 + a^2 + ab - 1 > 0$$

$$a^2 + ab + 1 > 0$$

its discriminant is negative

$$D = b^2 - 4 < 0 = -2 < b < 2$$

No. of integral value of 'b' is $\{-1, 0, 1\}$

20. **Ans (A)**

We have $\sim (p \vee q) \vee (\sim p \wedge q)$

$$(\sim p \wedge \sim q) \vee (\sim p \wedge q)$$

$$\sim p \wedge (\sim q \vee q) = \sim p$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (2022)**

$$x^4 - 4x^3 + 6x^2 - 4x + 1 = 2022$$

$$(x-1)^4 = 2022$$

$(x-1)^2 = -\sqrt{2022}$ because only non real roots required

$$x^2 + 1 - 2x + \sqrt{2022} = 0$$

$$x^2 - 2x + 1 + \sqrt{2022} = 0$$

$$p = \text{product of roots} = 1 + \sqrt{2022}$$

2. **Ans (0)**

Equation of normal at P ($3 \cos \theta, 2 \sin \theta$) is

$$3x \sec \theta - 2y \csc \theta = 5$$

Now it is tangent to circle so

$$\frac{5}{\sqrt{9 \sec^2 \theta + 4 \csc^2 \theta}} = \sqrt{3}$$

$$\text{But } \{9 \sec^2 \theta + 4 \csc^2 \theta\} \geq (3+2)^2 = 25$$

No such θ exist

3. **Ans (0)**

$$\sin x \cdot \sin\left(\frac{1}{x}\right) = 1$$

$$\sin\left(\frac{1}{x}\right) = \csc x$$

$$\text{Clearly } \sin\left(\frac{1}{x}\right) = \csc x = 1 \text{ or}$$

$$\sin\left(\frac{1}{x}\right) = \csc x = -1 \text{ which is not possible}$$

for same value of x.

4. **Ans (1)**

$$x \in \text{prime and } x < 10 \Rightarrow x = 2, 3, 5, 7$$

Total no. of A(x, y) pair $n(A) = 4 \times 10 = 40$ and

$$x^2 - 3y^2 = 1 \Rightarrow x^2 = 3y^2 + 1.$$

For above condition to be satisfied only two such pairs (2, 1) and (7, 4) are possible

$$\Rightarrow P(A) = \frac{2}{40} = \frac{1}{20} = P$$

5. **Ans (2)**

$$\begin{aligned} \lim_{k \rightarrow \infty} \int_0^{k[x]} (\{kt\})^k &= \lim_{k \rightarrow \infty} \int_0^{k^2[x] \cdot 1/k} (\{kt\})^k \\ &= \lim_{k \rightarrow \infty} k^2[x] \int_0^{1/k} (kt)^k \cdot dt \\ &= \lim_{k \rightarrow \infty} k^2[x] \cdot k^k \cdot \left[\frac{t^{k+1}}{k+1} \right]_0^{1/k} \\ &= \lim_{k \rightarrow \infty} k^2[x] \cdot k^k \cdot \frac{1}{k^{k+1}(k+1)} \\ &= [x] \lim_{k \rightarrow \infty} \frac{k}{k+1} = [x] \text{ or } \lambda = 2 \end{aligned}$$

6. **Ans (9)**

$$\begin{aligned} \lim_{x \rightarrow 0} \left[\frac{\sin^{-1} x}{\sin x} \right] &= 1 \quad 1 + 2 + 3 + \dots + n = 45 \\ \lim_{x \rightarrow 0} \left[\frac{\sin^{-1} 2x}{\sin x} \right] &= 2 \quad \frac{n(n+1)}{2} = 45 \\ \lim_{x \rightarrow 0} \left[\frac{\sin^{-1} 3x}{\sin x} \right] &= 3 \quad n(n+1) = 90 \\ \lim_{x \rightarrow 0} \left[\frac{\sin^{-1} nx}{\sin x} \right] &= n \quad n = 9 \end{aligned}$$

7. **Ans (5)**

$$\begin{aligned} T_r &= \cot^{-1} \left(4 + \frac{r(r+1)}{4} \right) \\ &= \tan^{-1} \left(\frac{1}{4 + \frac{r(r+1)}{4}} \right) \\ &= \tan^{-1} \left(\frac{1}{4 \left(1 + \frac{r(r+1)}{16} \right)} \right) \\ &= \tan^{-1} \left(\frac{1/4}{1 + \frac{r}{4} \cdot \frac{(r+1)}{4}} \right) \\ &= \tan^{-1} \left(\frac{\frac{r+1}{4} - \frac{r}{4}}{1 + \frac{r}{4} \cdot \frac{r+1}{4}} \right) \\ T_r &= \tan^{-1} \left(\frac{r+1}{4} \right) - \tan^{-1} \left(\frac{r}{4} \right) \\ S_n &= \sum_{r=1}^n T_r = \tan^{-1} \left(\frac{n+1}{4} \right) - \tan^{-1} \left(\frac{1}{4} \right) \\ S_\infty &= \tan^{-1}(\infty) - \tan^{-1} \left(\frac{1}{4} \right) \\ &= \frac{\pi}{2} - \tan^{-1} \left(\frac{1}{4} \right) = \cot^{-1} \left(\frac{1}{4} \right) = \tan^{-1}(4) \\ \text{So } a &= 4, b = 1 \end{aligned}$$

8. **Ans (11)**

Rearrangement of 5 things

$$\begin{aligned} 5! \left(\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) \\ \frac{5!}{2!} - \frac{5!}{3!} + \frac{5!}{4!} - \frac{5!}{5!} \\ = 60 - 20 + 5 - 1 = 44 \end{aligned}$$

Now $f(1)$ can be 2, 3, 4, 5 in above arrangement

if $f(1)$ has to be '2' then arrangements will be

$$\frac{44}{4} = 11$$

9. **Ans (5)**

From 7! every number will be divisible by 14

so $1! + 2! + 3! + 4! + 5! + 6! = 873$

Now remainder when it is divided by 14 is '5'.

10. **Ans (22)**

Use LMVT in $[1, 2]$

$$\frac{f(2) - f(1)}{2 - 1} = f'(C) \geq 1$$

$$f(2) - 3 \geq 1$$

$$f(2) \geq 4$$

Now use LMVT in $[2, 4]$

$$\frac{f(4) - f(2)}{4 - 2} = f'(C_1) \geq 1$$

$$f(4) - f(2) \geq 2$$

$$f(2) \leq 7$$

$$4 \leq f(2) \leq 7$$

Possible integral values 4, 5, 6, 7

sum of values $4 + 5 + 6 + 7 = 22$